Algorithm 1 Pose-Aware Channel Importance Pruning

Input: Pretrained HRNet-W48 model $\mathcal{M}_0$, calibration subset $\mathcal{D}_{\text{calib}} \subset \mathcal{D}$, branch pruning budgets ρ_{high}, ρ_{low}, target pruning ratio R_{target}

Output: Structurally pruned model $\mathcal{M}_{\text{pruned}}$, optimal pruning rate r^*

$\mathcal{M} \leftarrow \mathcal{M}_0$
▷ Step 1: Task-driven channel importance assessment
for each convolutional channel c in $\mathcal{M}$ **do**
    $\mathbf{H}_c \leftarrow \mathcal{M}(x \mid \text{channel } c \leftarrow 0)$ on $\mathcal{D}_{\text{calib}}$
    $\varepsilon_c \leftarrow \|\mathbf{H}_c - y\|^2$
    $\varepsilon_0 \leftarrow \|\mathbf{H}_0 - y\|^2$ ▷ baseline error
    $s(c) \leftarrow \varepsilon_c - \varepsilon_0$ ▷ importance score: error gain
end for
▷ Step 2: Branch-differentiated channel retention
for each branch $b \in \{\text{high-res}, \text{low-res}\}$ **do**
    $S_b \leftarrow \{s(c) \text{ for channels } c \text{ in branch } b\}$
    if b is high-resolution branch **then**
        $\text{keep\_ratio}_b \leftarrow 1 - \rho_{\text{high}}$ ▷ conservative budget
    else
        $\text{keep\_ratio}_b \leftarrow 1 - \rho_{\text{low}}$ ▷ aggressive budget
    end if
    $k_b \leftarrow \lceil |S_b| \times \text{keep\_ratio}_b \rceil$
    Retain top k_b channels in branch b sorted by S_b
end for
▷ Step 3: Golden-section search for optimal pruning rate
Initialize interval $[\alpha, \beta] \leftarrow [0.3, 0.8]$
while $\beta - \alpha > \epsilon$ **do**
    $r_1 \leftarrow \alpha + 0.382(\beta - \alpha)$
    $r_2 \leftarrow \alpha + 0.618(\beta - \alpha)$
    Evaluate $\mathcal{F}(r) = \frac{\text{Acc}(r)}{\text{Acc}_{\max}} + \frac{\text{AP}(r)}{\text{AP}_{\max}} + r$
    if $\mathcal{F}(r_1) > \mathcal{F}(r_2)$ **then**
        $\beta \leftarrow r_2$
    else
        $\alpha \leftarrow r_1$
    end if
end while
$r^* \leftarrow (\alpha + \beta)/2$
$\mathcal{M}_{\text{pruned}} \leftarrow$ Remove channels with lowest scores to reach r^*
if $r^* < R_{\text{target}}$ **then**
    Adjust ρ_{high}, ρ_{low} and repeat from Step 2
end if
return $\mathcal{M}_{\text{pruned}}$, r^*

Algorithm 2 Pruning-Aware Dynamic Distillation Weight Scheduling

Input: Structurally pruned model $\mathcal{M}_{\text{pruned}}$, PoseBH teacher model $\mathcal{T}$, training set $\mathcal{D} = \{(x_i, y_i)\}$, target pruning ratio R_{target}, number of pruning steps N, epochs per step E

Output: Compact distilled model $\mathcal{M}^*$

$\triangleright$ Initialize dynamic scheduling parameters
$\lambda_{\text{schedule}} \leftarrow [\lambda_{\text{high}}, \lambda_{\text{mid}}, \lambda_{\text{low}}]$
$\alpha_{\text{schedule}} \leftarrow [1.0, 0.5, 0.0]$ $\triangleright$ 1: global, 0: local
$\triangleright$ Progressive pruning with adaptive distillation
for step $= 1$ to N **do**
    Progressively prune $\mathcal{M}_{\text{pruned}}$ toward R_{target}
    $\triangleright$ Determine pruning stage
    if step $\leq N/3$ **then**
        stage $\leftarrow$ structural stabilization
    else if step $\leq 2N/3$ **then**
        stage $\leftarrow$ accuracy recovery
    else
        stage $\leftarrow$ fine-tuning
    end if
    $\lambda \leftarrow \lambda_{\text{schedule}}[\text{stage}]$
    $\alpha \leftarrow \alpha_{\text{schedule}}[\text{stage}]$
    $\triangleright$ Distillation-guided fine-tuning
    for epoch $= 1$ to E **do**
        for each mini-batch $(x, y) \in \mathcal{D}$ **do**
            $\mathbf{H}_s, \mathbf{F}_s \leftarrow \mathcal{M}_{\text{pruned}}(x)$
            $\mathbf{H}_t, \mathbf{F}_t \leftarrow \mathcal{T}(x)$
            $\mathcal{L}_{\text{task}} \leftarrow \text{MSE}(\mathbf{H}_s, y)$
            $\mathcal{L}_{\text{global}} \leftarrow \|\mathbf{F}_s - \mathbf{F}_t\|^2$
            $\mathcal{L}_{\text{local}} \leftarrow \text{KL}(\mathbf{H}_s \| \mathbf{H}_t)$
            $\mathcal{L} \leftarrow \mathcal{L}_{\text{task}} + \lambda\big[\alpha \cdot \mathcal{L}_{\text{global}} + (1 - \alpha) \cdot \mathcal{L}_{\text{local}}\big]$
            Update $\mathcal{M}_{\text{pruned}}$ via AdamW optimizer
        end for
    end for
end for
return $\mathcal{M}^* \leftarrow \mathcal{M}_{\text{pruned}}$